

Experiment No.

Date:

KICAD Individual Project

Objective:

1. To design the schematic layout.
2. To simulate the schematic for verification.
3. To design the PCB layout.
4. To generate Gerber and Excellon drill files.

Equipment:

Sr. No	Resources
1.	Desktop/Laptop System
2.	External Mouse
3.	KiCAD

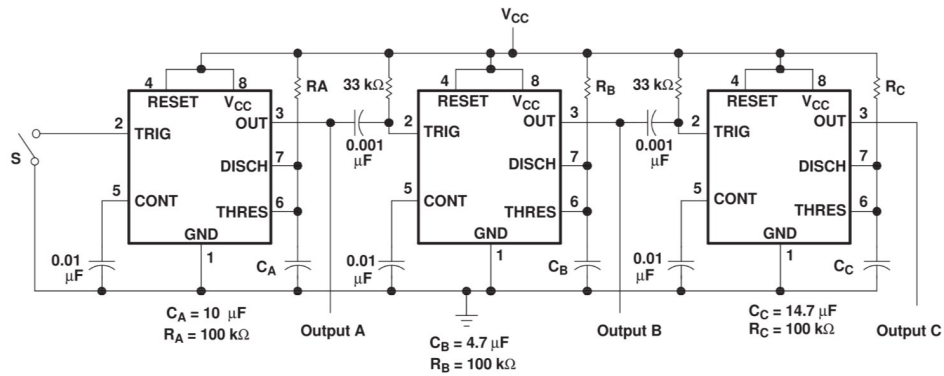
Procedure:

1. Study the working of the circuits, download the component datasheets, and find the dimensions of the components.
2. Design the schematic layout:
 - a. Open Schematic Editor. Set grid and page settings.
 - b. Add symbols to the sheet according to the circuit.
 - c. Arrange, wire, and annotate symbols; add textual information ensuring readability.
 - d. Perform Electrical Rules Check (ERC), fix errors, and repeat as necessary.
 - e. Assign footprints for each symbol according to finalized datasheets.
 - f. Attach datasheets to the schematic symbols.
 - g. Export the Bill of Materials (BOM) file.
 - h. Export the netlist.
3. Design the PCB layout:
 - a. Open PCB Layout Editor. Set grid and page settings.
 - b. Import the generated netlist.
 - c. Arrange footprints by functional blocks and user interface needs, ensuring airwires do not intersect and the 'Ratsnest' is resolved.
 - d. Route the tracks; add textual info (your name, PCB type abbreviation) ensuring readability; avoid acute angles in tracks to prevent acid traps during etching.
 - e. Add the edge cuts layer defining PCB boundaries; keep PCB size minimal within constraints.
 - f. Perform zone fill connecting ground terminals.
 - g. Perform Design Rules Check (DRC), fix errors, and repeat.
 - h. Create, verify, and export Gerber and Excellon drill files.
4. Perform a simulation:
 - a. Search and assign suitable SPICE models for unassigned components.
 - b. Open the simulator.
 - c. Select the signals whose waveforms you wish to view from the available list.
 - d. Click the Run button.



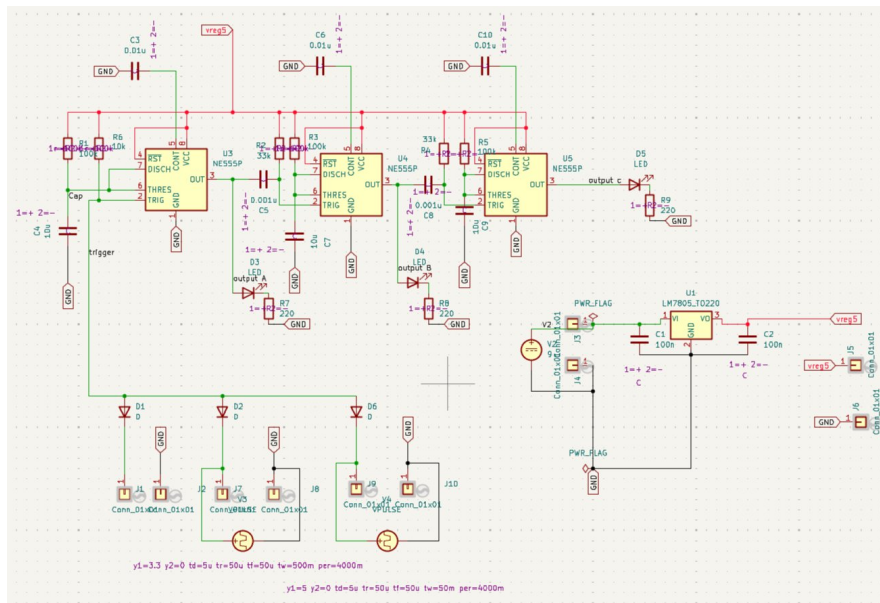
Circuit Diagram:

IC555 based sequential timer circuit



Implementation:

Sequential Timer Schematic Circuit:

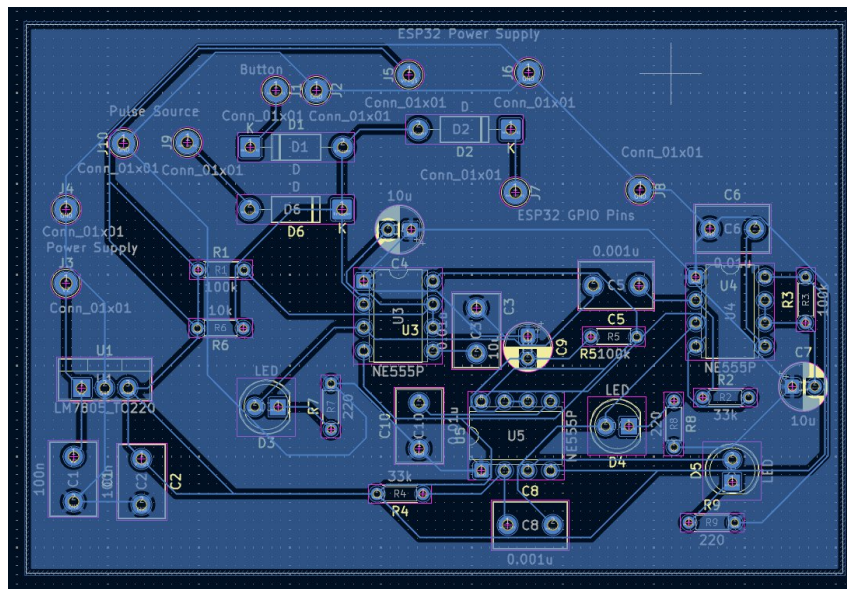


Schematic Circuit after ERC

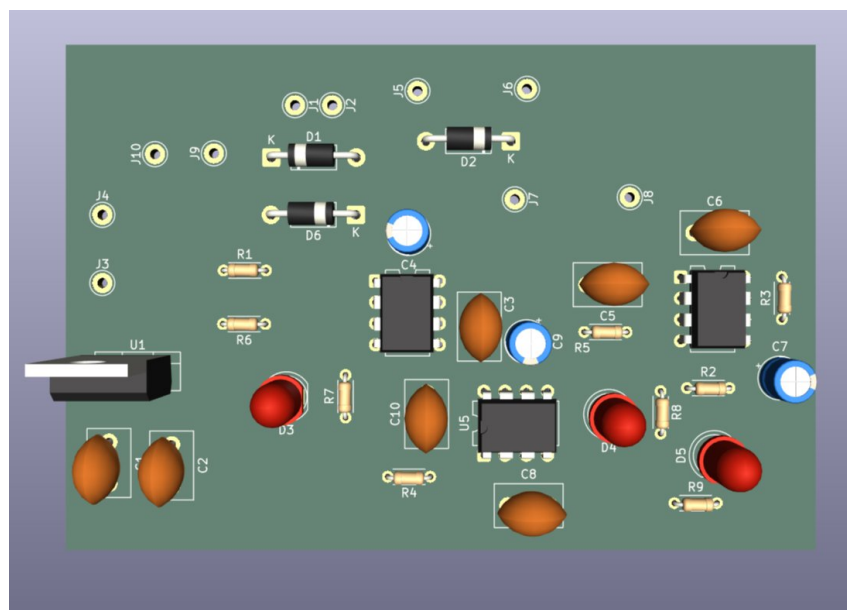
Bill of Materials:

Reference	Qty	Value	DNP	Exclude from BOM	Exclude from Board	Footprint
► C1, C2	2	100n	☐	☐	☐	Capacitor_THT:C_Disc_D8.0mm_W5.0mm_P5.00mm
► C3, C6, C10	3	0.01u	☐	☐	☐	Capacitor_THT:C_Disc_D8.0mm_W5.0mm_P5.00mm
► C4, C9	2	10u	☐	☐	☐	Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
► C5, C8	2	0.001u	☐	☐	☐	Capacitor_THT:C_Disc_D8.0mm_W5.0mm_P5.00mm
C7	1	10u	☐	☒	☐	Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
► D1, D2, D6	3	D	☐	☐	☐	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal
► D3-D5	3	LED	☐	☐	☐	LED_THT:LED_D5.0mm
► J1-J10	10	Conn_01x01	☐	☐	☐	Connector_Pin:Pin_D1.0mm_L10.0mm_LooseFit
► R1, R3, R5	3	100k	☐	☐	☐	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal
► R2, R4	2	33k	☐	☐	☐	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal
R6	1	10k	☐	☐	☐	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal
► R7-R9	3	220	☐	☐	☐	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal
U1	1	LM7805_TO220	☐	☐	☐	Package_TO_SOT_THT:TO-220-3_Vertical
► U3-U5	3	NE555P	☐	☐	☐	Package_DIP:DIP-8_W7.62mm
V2	1	9	☐	☐	☒	
► V3, V4	2	VPULSE	☐	☐	☒	

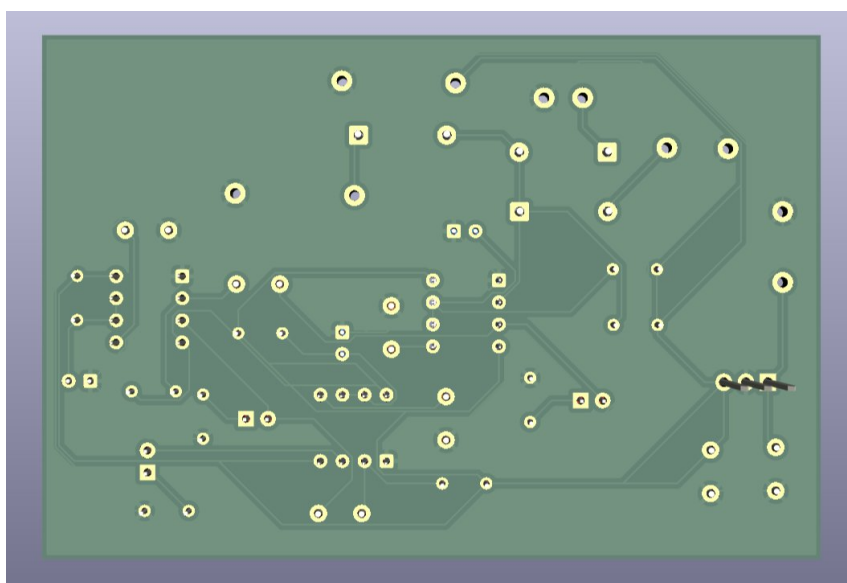




PCB Layout after DRC

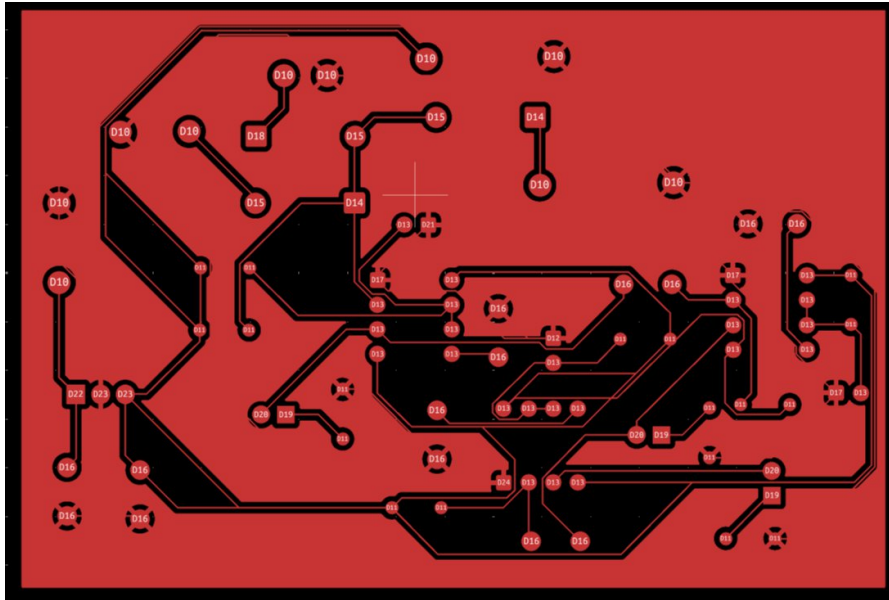


Component-side 3D view

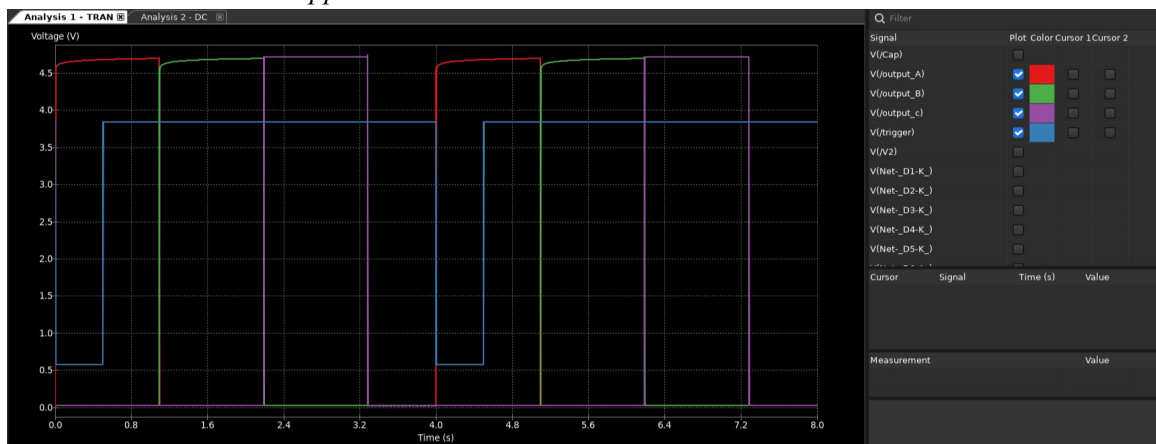


Solder-side 3D view





Back Copper Gerber and Excellon Drill Files in Gerber Viewer



Output Waveform

Challenges:

1. Finding and selecting the right components for the design.
2. Finding the right SPICE model associated with IC555.
3. Assigning the correct footprints to the components.
4. Identifying and fixing errors reported during the Electrical Rules Check (ERC).
5. Avoiding overlaps and congestion during PCB layout to ensure clear routing.
6. Maintaining proper synchronization between the schematic and PCB layout.

Conclusion:

1. PCB designing process was studied and practiced.
2. The schematic and PCB layout were designed and verified using ERC (Electrical Rules Check) and DRC (Design Rules Check).
3. IC555 based sequential timer output was observed and analyzed.

